

McDiarmid's Inequality for Local Stochastic Noise

David Ponarovsky

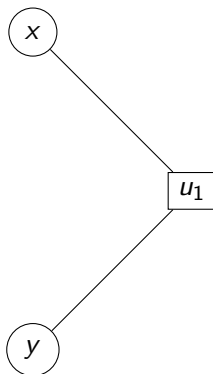
December 2, 2025

Introduction

Today:

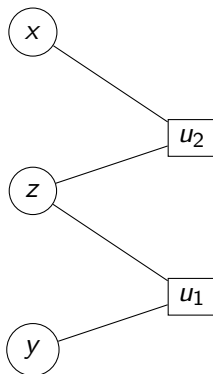
- ▶ McDiarmid's Inequality for Local Stochastic Noise.
- ▶ Dependence.
- ▶ Local Stochastic Noise.
- ▶ IQP.

Dependence Type I



$$\begin{aligned} & \mathbf{Pr}[(x, y) \in e | u_1 \text{ is satisfied}] \\ &= \mathbf{Pr}[x \in e | u_1 \text{ is satisfied}] \mathbf{Pr}[y \in e | u_1 \text{ is satisfied}] \\ &\leq \mathbf{Pr}[x \in e] \mathbf{Pr}[y \in e] \end{aligned}$$

Dependence Type II



$$\begin{aligned} & \mathbf{Pr}[(x, y) \in e | z \text{ is not flipped}] \\ &= \mathbf{Pr}[x \in e | z \text{ is not flipped}] \mathbf{Pr}[y \in e | z \text{ is not flipped}] \\ &\leq \mathbf{Pr}[x \in e] \mathbf{Pr}[y \in e] \end{aligned}$$

Definition

Consider two (qu)bits x, y . We say that z is a **linked node** between x and y if z is either a bit or a check that is shared by both of their D light cones.

Remark

Notice that y might be a **linked node** between x and y .

Definition

Consider an application of a noise channel. We say that z is a **clean linked node** if:

1. z is a bit and $z \notin f$ (not flipped).
2. z is a check and z is satisfied.

$$\begin{aligned}
& \Pr[x_1, x_2, \dots, x_k, x_{k+1}, \dots, x_l \in e \mid z_1, z_2, \dots, z_m] \\
&= \prod_i \Pr[x_i \in e \mid z_1, z_2, \dots, z_m] \\
&= \prod_i \Pr[x_i \in e \mid z \in \Gamma(x_i)] \\
&\leq \prod_{i=1}^k \Pr[x_i \in e] \prod_{i=k+1}^l \Pr[x_i \in e \mid z_i \in \Gamma(x_i)] \\
&\leq M(p)^k
\end{aligned}$$

McDiarmid's Inequality - Local Stochastic Noise

C_1, C_2 - local stochastic noise

Let $C_1, C_2 > 0$. We say that ρ is subjected to C_1, C_2 - local stochastic noise, if for any subset of qubits S the probability they will be faulty is bounded by:

$$C_1 p^{|S|} \leq \Pr[S \in e] \leq C_2 p^{|S|}$$

Lemma: McDiarmid's Inequality

McDiarmid's Inequality is correct for C_1, C_2 - local stochastic noise.

Proof

Proof

Consider ref. We have to bound:

$$\star = \sup_{x, \bar{x}} \mathbf{E} [f | X_1, \dots, X_{i-1}, x] - \mathbf{E} [f | X_1, \dots, X_{i-1}, \bar{x}]$$

First we show that:

$$\begin{aligned} \frac{C_2}{C_1} p^{|Z|} &\leq \mathbf{Pr} [z_{i+1}, \dots, z_n | X_1, \dots, X_{i-1}, x] \leq \frac{C_2}{C_1} p^{|Z|} \\ \frac{C_2}{C_1} p^{|Z|} &\leq \mathbf{Pr} [z_{i+1}, \dots, z_n | X_1, \dots, X_{i-1}, \bar{x}] \leq \frac{C_2}{C_1} p^{|Z|} \end{aligned}$$

Proof

And then we get that:

$$\begin{aligned}\star &\leq \sum_Z p^{|Z|} \left(\frac{C_2}{C_1} - \frac{C_1}{C_2} \right) c \\ &\leq c \left(\frac{C_2}{C_1} - \frac{C_1}{C_2} \right) \frac{1}{1-p}\end{aligned}$$

Trick.

Suppose that $\mathbf{Pr}[S]$ is not bounded from beneath by $C_1 p^{|S|}$. Then one can define the distribution $\mathcal{D}'$ such that:

$$\mathcal{D}'(S) = \begin{cases} \mathcal{D}(S) & \text{if } \mathcal{D}(S) > C_1 p^{|S|} \\ C_1 p^{|S|} & \text{else} \end{cases}$$

Normalize by:¹

$$\begin{aligned} \xi &= \sum_{\mathcal{D}(S) < C_1 p^{|S|}} C_1 p^{|S|} + \mathbf{Pr}[\mathcal{D}(S) > C_1 p^{|S|}] \\ &\geq \begin{cases} \sum_S p^{|S|} \geq 1 & \text{if } C_1 < 1 \\ C_1 \sum_S p^{|S|} \geq C_1 & \text{else} \end{cases} \\ &\geq C_1 \end{aligned}$$

¹if $C_1 < 1$

Trick.

So for $C_1 > 1$ we have:

$$\begin{aligned}\mathbf{E}_{\sim \mathcal{D}'}[f] &= \frac{1}{\xi} \sum_{\Pr[\mathcal{D}(S) < C_1 p^{|S|}]} f(S) C_1 p^{|S|} + \sum_{\Pr[\mathcal{D}(S) \geq C_1 p^{|S|}]} f(S) D(S) \\ &\leq \frac{1}{\xi} \mathbf{E}_{\sim \mathcal{D}}[f] \\ &\leq \mathbf{E}_{\sim \mathcal{D}}[f]\end{aligned}$$

Trick.

Then:

$$\begin{aligned} & \Pr_{\sim \mathcal{D}} [f - \mathbf{E}_{\sim \mathcal{D}} [f] > t] \\ & \leq \Pr_{\sim \mathcal{D}' \text{ not normalized}} [f - \mathbf{E}_{\sim \mathcal{D}} [f] > t] \\ & \leq \frac{1}{C_1} \Pr_{\sim \mathcal{D}'} [f - \mathbf{E}_{\sim \mathcal{D}} [f] > t] \\ & \leq \Pr_{\sim \mathcal{D}'} [f - \mathbf{E}_{\sim \mathcal{D}} [f] > t] \\ & \leq \Pr_{\sim \mathcal{D}'} [f - \mathbf{E}_{\sim \mathcal{D}'} [f] > t] \end{aligned}$$

Local Decoder

definition

We say that D is a local decoder if when applied against $2p(1-p)$ -depolarized channel:

- ▶ There exists function $M : (0, 1) \rightarrow (0, 1)$ such that the probability of each bit being flipped is $M(2p(1-p)) < p^{\frac{100}{99}}$.
- ▶ There is a constant l such that for any bit, there are only $(\Delta)^l$ bits on which the event of it being flipped depends.

For example, if we take k large enough and consider the (n, k) -Toric, then both the swift-rule and the flip upon majority are local decoders.

Definition

Consider the moment after the application of the depolarizing channel, and right before the correction by the decoder.

For a subset of qubits S , we say that a qubit $x \in S$ is depended if there exists $y \in S$ such:

1. Either x and y share a check that is not satisfied. (Type I)
2. Or, x and y have checks which share a bit that was flipped by the application of the channel. (Type II)